# ENTERPRISE NETWORK DESIGN AND TROUBLESHOOTING LAB

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Technology: Cisco Packet Tracer

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Purpose: Macquarie Network Engineer Portfolio Project

## 1. PROJECT OVERVIEW

### Objective

Design, implement, secure, and troubleshoot a three-tier enterprise network simulating a real corporate environment with three departments — HR, Finance, and IT. This project directly demonstrates skills required for enterprise network engineer roles including routing and switching, VLAN configuration, access control, dynamic routing, NAT, and systematic fault resolution.

### Technologies Used

- Cisco VLANs and 802.1q trunking
- Router on a Stick inter-VLAN routing
- OSPF dynamic routing protocol
- NAT overload for internet access
- Extended ACLs for department access control
- Network fault simulation and troubleshooting
- Cisco IOS show commands for diagnosis

## Network Design Decisions

Router on a Stick was chosen over Layer 3 switch inter-VLAN routing to ensure ACL enforcement on all inter-department traffic. By routing all VLAN traffic through physical router subinterfaces, access control policies are applied consistently to every packet regardless of source or destination department.

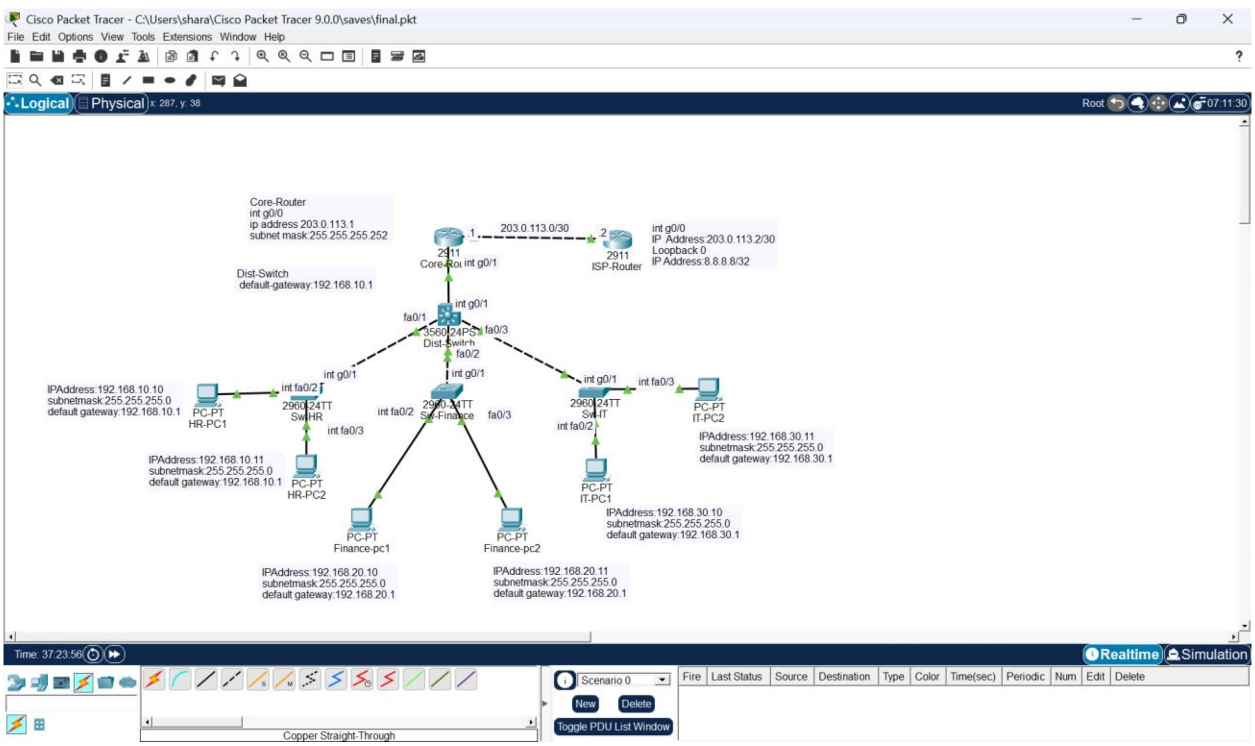
# 2. NETWORK TOPOLOGY

The network follows a three-tier hierarchical design commonly used in enterprise environments:

**Tier 1 — Core:** ISP-Router and Core-Router handling internet connectivity, NAT, and security policy

**Tier 2 — Distribution:** Dist-Switch (3560) aggregating all access switches via trunk links

**Tier 3 — Access:** SW-HR, SW-Finance, SW-IT connecting end user PCs to the network




## Device Inventory

| Device | Model | Role | Location |
|---|---|---|---|
| ISP-Router | Cisco 2911 | Simulates internet ISP | Edge |
| Core-Router | Cisco 2911 | NAT, ACL, inter-VLAN routing | Core |
| Dist-Switch | Cisco 3560 | Aggregation, trunking | Distribution |
| SW-HR | Cisco 2960 | HR department access | Access |
| SW-Finance | Cisco 2960 | Finance department access | Access |
| SW-IT | Cisco 2960 | IT department access | Access |
| HR-PC1, HR-PC2 | PC-PT | HR end users | HR Dept |
| Finance-PC1, PC2 | PC-PT | Finance end users | Finance Dept |
| IT-PC1, IT-PC2 | PC-PT | IT end users | IT Dept |

## 3. IP ADDRESSING TABLE

| Device | Interface | IP Address | Subnet Mask | Gateway |
|---|---|---|---|---|
| ISP-Router | Gi0/0 | 203.0.113.2 | 255.255.255.252 | N/A |
| ISP-Router | Loopback0 | 8.8.8.8 | 255.255.255.255 | N/A |
| Core-Router | Gi0/0 | 203.0.113.1 | 255.255.255.252 | N/A |
| Core-Router | Gi0/1.10 | 192.168.10.1 | 255.255.255.0 | N/A |
| Core-Router | Gi0/1.20 | 192.168.20.1 | 255.255.255.0 | N/A |
| Core-Router | Gi0/1.30 | 192.168.30.1 | 255.255.255.0 | N/A |
| HR-PC1 | Fa0 | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 |
| HR-PC2 | Fa0 | 192.168.10.11 | 255.255.255.0 | 192.168.10.1 |
| Finance-PC1 | Fa0 | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 |
| Finance-PC2 | Fa0 | 192.168.20.11 | 255.255.255.0 | 192.168.20.1 |

| | | | | |
|---|---|---|---|---|
| IT-PC1 | Fa0 | 192.168.30.10 | 255.255.255.0 | 192.168.30.1 |
| IT-PC2 | Fa0 | 192.168.30.11 | 255.255.255.0 | 192.168.30.1 |

# 4. VLAN CONFIGURATION

| VLAN ID | Name | Subnet | Gateway | Devices |
|---|---|---|---|---|
| 10 | HR | 192.168.10.0/24 | 192.168.10.1 | HR-PC1, HR-PC2 |
| 20 | Finance | 192.168.20.0/24 | 192.168.20.1 | Finance-PC1, PC2 |
| 30 | IT | 192.168.30.0/24 | 192.168.30.1 | IT-PC1, IT-PC2 |

## Trunk Links

| From | To | Encapsulation | VLANs Allowed |
|---|---|---|---|
| Core-Router Gi0/1 | Dist-Switch Gi0/1 | 802.1q | 10, 20, 30 |
| Dist-Switch Fa0/1 | SW-HR Gi0/1 | 802.1q | 10 |
| Dist-Switch Fa0/2 | SW-Finance Gi0/1 | 802.1q | 20 |
| Dist-Switch Fa0/3 | SW-IT Gi0/1 | 802.1q | 30 |

# 5. SECURITY POLICY — ACL CONFIGURATION

Extended ACL named DEPT-POLICY was implemented on Core-Router to enforce department access control. The ACL is applied inbound on all three subinterfaces (Gi0/1.10, Gi0/1.20, Gi0/1.30) ensuring all inter-department traffic is inspected.

## ACL Rules

| Rule | Source | Destination | Action | Reason |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 10 | 192.168.30.0/24 (IT) | Any | PERMIT | IT manages all depts |
| 20 | Any | 192.168.30.0/24 (IT) | PERMIT | Allow return traffic to IT |
| 30 | 192.168.20.0/24 (Finance) | 192.168.10.0/24 (HR) | DENY | Protect HR data |
| 40 | 192.168.10.0/24 (HR) | 192.168.20.0/24 (Finance) | DENY | Protect Finance data |
| 50 | Any | Any | PERMIT | Allow all other traffic |

### Verification Results

| Source | Destination | Result | Status |
|---|---|---|---|
| Finance-PC1 (192.168.20.10) | HR-PC1 (192.168.10.10) | BLOCKED | ✓ Policy enforced |
| HR-PC1 (192.168.10.10) | Finance-PC1 (192.168.20.10) | BLOCKED | ✓ Policy enforced |
| IT-PC1 (192.168.30.10) | HR-PC1 (192.168.10.10) | ALLOWED | ✓ IT access confirmed |
| IT-PC1 (192.168.30.10) | Finance-PC1 (192.168.20.10) | ALLOWED | ✓ IT access confirmed |
| All PCs | 8.8.8.8 (Internet) | ALLOWED | ✓ NAT working |

# 6. PHASE 4 — FAULT SIMULATION AND TROUBLESHOOTING

Five real-world network faults were deliberately introduced and resolved using systematic troubleshooting methodology. Each fault simulates a common enterprise network incident.

### Fault 1 — Trunk Link Failure

**Fault Introduced:**

```
shutdown — interface FastEthernet0/1 on Dist-Switch
```

**Symptom Observed:**

HR-PC1 lost all network connectivity. ping 192.168.10.1 — request timed out

**Diagnosis Commands:**

```
show interfaces FastEthernet0/1  →  administratively down detected
show interfaces trunk  →  Fa0/1 missing from trunk list
show vlan brief  →  VLAN 10 showing no active ports
```

**Fix Applied:**

```
no shutdown — interface FastEthernet0/1 on Dist-Switch
```

**Verification:**

show interfaces trunk confirmed Fa0/1 trunking. ping 192.168.10.1 from HR-PC1 successful

### Fault 2 — VLAN Misconfiguration

**Fault Introduced:**

```
switchport access vlan 99 — interface FastEthernet0/2 on SW-Finance
```

**Symptom Observed:**

Finance-PC1 lost all connectivity. ping 192.168.20.1 — request timed out

**Diagnosis Commands:**

```
show vlan brief  →  Fa0/2 moved to nonexistent VLAN 99
VLAN 20 showing only Fa0/3 — Fa0/2 missing
```

**Fix Applied:**

```
switchport access vlan 20 — interface FastEthernet0/2 on SW-Finance
```

**Verification:**

show vlan brief confirmed Fa0/2 and Fa0/3 under VLAN 20. ping 192.168.20.1 successful

### Fault 3 — Router Interface Failure

**Fault Introduced:**

```
shutdown — interface GigabitEthernet0/1 on Core-Router
```

**Symptom Observed:**

All PCs lost complete network connectivity. ping 8.8.8.8 — request timed out

**Diagnosis Commands:**

```
show ip interface brief  →  Gi0/1 administratively down
show ip route  →  all subinterfaces down, O routes missing
Gateway of last resort not set
```

**Fix Applied:**

```
no shutdown — interface GigabitEthernet0/1 on Core-Router
```

**Verification:**

show ip interface brief confirmed Gi0/1 up/up. All O routes restored. ping 8.8.8.8 successful

### Fault 4 — OSPF Network Statement Removed

**Fault Introduced:**

```
no network 172.16.0.0 0.0.0.3 area 0 — router ospf 1 on Core-Router
```

**Symptom Observed:**

All PCs lost internet access. ping 8.8.8.8 — request timed out

**Diagnosis Commands:**

```
show ip ospf neighbor  →  blank output, no neighbours
show ip route  →  no O routes, gateway of last resort not set
show running-config | section ospf  →  network statement missing
```

**Fix Applied:**

```
network 172.16.0.0 0.0.0.3 area 0 — router ospf 1 on Core-Router
```

**Verification:**

show ip ospf neighbor confirmed FULL state. O routes restored. ping 8.8.8.8 successful

### Fault 5 — Wrong Default Gateway on PC

**Fault Introduced:**

```
Default gateway changed from 192.168.30.1 to 192.168.30.99 on IT-PC1
```

**Symptom Observed:**

IT-PC1 lost all inter-subnet connectivity. Same subnet ping worked. ping 8.8.8.8 failed

**Diagnosis Commands:**

```
Observed pattern: same subnet works, outside subnet fails
This pattern immediately indicates gateway misconfiguration
Checked IT-PC1 IP Configuration — gateway 192.168.30.99 confirmed incorrect
```

**Fix Applied:**

```
Default gateway corrected to 192.168.30.1 on IT-PC1 IP Configuration
```

**Verification:**

ping 8.8.8.8 successful. ping 192.168.10.10 successful. Full connectivity restored

# 7. LESSONS LEARNED

## 1. Router on a Stick vs Layer 3 Switching

Initially configured inter-VLAN routing using Layer 3 switching on the Dist-Switch with SVIs. Discovered that Packet Tracer has limitations enforcing ACLs on virtual SVI interfaces. Redesigned using Router on a Stick which forces all inter-VLAN traffic through physical router subinterfaces enabling reliable ACL enforcement on every packet.

### 2. ACL Placement is Critical

ACLs must be placed where traffic physically passes through. Traffic routed internally by a Layer 3 switch can bypass router ACLs entirely. Understanding traffic flow paths is essential before applying access control policies.

### 3. OSPF Dependency Chain

Removing a single OSPF network statement caused complete loss of internet access for all departments. This demonstrated how critical dynamic routing configuration is and why changes should always be verified with show ip ospf neighbor and show ip route.

### 4. Systematic Troubleshooting

The most effective troubleshooting approach was working layer by layer — physical first (show interfaces), then data link (show vlan brief, show interfaces trunk), then network (show ip route, show ip ospf neighbor). This OSI model approach consistently identified root causes efficiently.

### 5. Gateway Misconfiguration Pattern

A wrong default gateway causes loss of all inter-subnet connectivity while same-subnet communication still works. Recognising this symptom pattern immediately points to a gateway issue without needing extensive diagnosis.

## 8. CONCLUSION

This project successfully demonstrated the design, implementation, security, and troubleshooting of a three-tier enterprise network. All core network engineering skills were applied and verified including VLAN segmentation, dynamic routing, internet access via NAT, access control enforcement, and systematic fault resolution.

The skills demonstrated directly align with enterprise network engineering requirements and provide a practical foundation for roles requiring hands-on Cisco IOS experience, network troubleshooting capability, and understanding of enterprise security architecture.

### Future Enhancements

- Add redundant trunk links with STP verification and failover testing
- Implement DHCP server for dynamic IP assignment across all VLANs
- Add wireless access points per department with separate SSIDs
- Configure port security on access switches to prevent unauthorised devices
- Implement QoS policies for voice and video traffic prioritisation
- Add a dedicated management VLAN for network device administration